



AI Playbook and Toolkit for Public Health Departments

AI Implementation Metrics and Value Tracking for Program Managers

PGM 130

This module helps program managers define and track whether AI-supported workflows create public health value. It focuses on practical metrics, burden, equity, quality, safety, adoption, cost, sustainability, and lifecycle decisions. The module emphasizes that value should be measured against public health purpose, not only speed, automation, or user enthusiasm.

Level	applied
Primary track	Program Management Track
Appears in tracks	program-management

Learning Objectives

- Define meaningful value measures for AI-supported public health workflows.
- Distinguish productivity metrics from public health value, equity, quality, and safety measures.
- Develop a monitoring plan that includes adoption, burden, errors, equity, and sustainment.
- Use implementation evidence to inform decisions to scale, revise, pause, or retire AI uses.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level training module is intended for public health education, planning, and workforce development. It does not replace agency policy, legal review, privacy review, security review, procurement review, communications clearance, civil rights review, Tribal consultation, or governance approval. Learners should confirm applicable requirements, approved tools, data rules, and review pathways within their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

AI projects are often justified by expected efficiency, but public health value is broader than time saved. A tool may reduce one task while increasing review burden, widening inequities, weakening trust, or creating new monitoring responsibilities. Program managers need to define what success means before implementation and track whether the workflow actually delivers value. This protects agencies from scaling tools that are impressive but not useful, safe, equitable, or sustainable.

Value tracking also supports governance. Governance bodies need evidence to decide whether an AI use should continue, change, expand, pause, or retire. Metrics should include process measures, quality measures, equity measures, user experience, risk indicators, and public health outcomes where feasible. A balanced approach helps teams avoid narrow success claims based only on speed or volume.

Public Health Example

A maternal and child health program pilots an AI-assisted referral follow-up prioritization process. The team tracks not only the number of referrals reviewed per day, but also whether high-priority families are contacted faster, whether staff spend more time correcting outputs, whether certain communities are under-prioritized, and whether families report clearer service navigation. The evidence shows that speed improved, but the prioritization logic missed families with incomplete data. The program revises the workflow before considering scale.

Technical, Programmatic, and Operational Deep Dive

A strong measurement plan begins with the public health purpose. The team should define the problem, the intended improvement, and the evidence that would show progress. Measures may include timeliness, completeness, accuracy, consistency, workload, user experience, service access, equity, safety, and cost. The plan should also identify measures that would show harm or unacceptable burden.

Productivity metrics should be interpreted carefully. Time saved may be meaningful, but only if quality and equity are maintained. A summarization tool that saves staff time but omits critical context is not valuable. A routing tool that processes more cases but misses low-data populations may worsen inequity. Program managers should pair efficiency measures with quality, safety, and equity indicators.

Value tracking should continue after go-live. AI-supported workflows can change over time because data, users, policies, populations, and vendor systems change. Monitoring should have owners, cadence, thresholds, and escalation pathways. The project record should show how evidence is used to improve the workflow and whether continuation remains justified.

Risks, Failure Modes, and Guardrails

A common failure mode is measuring only activity. Counts of messages processed, drafts generated, or cases flagged do not show whether the workflow improved public health practice. Guardrails include linking measures to the original use case purpose and including quality and equity measures. Teams should ask what better decision, service, communication, or outcome the AI use is supposed to support.

Another risk is ignoring hidden burden. AI may shift work to reviewers, supervisors, IT, data stewards, or community partners. A workflow that appears efficient for one team may create more work elsewhere. Burden measures should include time, cognitive load, rework, training, monitoring, and incident response. Staff feedback should be part of value tracking.

A third risk is failing to act on evidence. Monitoring has little value if findings are not reviewed or used. Governance and program leadership should define thresholds for revision, pause, or retirement. Evidence should support decisions rather than becoming a dashboard that no one uses.

Application to AI-Supported Workflows

This module applies to AI-supported workflows that are being piloted, implemented, monitored, or considered for scale. The measurement plan should be created before implementation and updated as the workflow matures. Program managers should coordinate with data, IT, equity, privacy, security, communications, and governance reviewers to ensure the measures reflect real risks and goals. Approved tools should be used for measurement and reporting when local policy requires them.

Reflection Questions

What would count as public health value for an AI-supported workflow in your program?

Which metrics could reveal hidden burden or inequitable impact?

Who should review the evidence and decide whether the AI use should continue?

Practical Exercise

Develop an AI implementation value tracking plan for one workflow. Include the public health purpose, efficiency measures, quality measures, equity measures, burden measures, safety indicators, data sources, owners, review cadence, thresholds, and lifecycle decision options.

Example: For AI-assisted call center FAQ support, the plan might track response time, accuracy of answers, staff corrections, accessibility of language, caller satisfaction, escalation rates, and whether certain language groups receive less accurate support.

Expected Artifact or Evidence

AI implementation value tracking plan.

Balanced measure set with at least one efficiency, quality, equity, burden, and safety measure.

Lifecycle decision criteria.

Knowledge Check

- 1. Why should AI value tracking include more than time saved?
- 2. What is a burden measure?
- 3. What should happen when monitoring identifies unacceptable harm?
- 4. Which measure set is most balanced?
- 5. When should value measures be defined?

References and Resources

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